

Ex: Fix a basis for $\mathbb{R}^2$ $\begin{bmatrix} x \\ y \end{bmatrix}$ 1) $4 \cdot 1 \cdot 1 + 1 \cdot 1 + 1 \cdot 1 + 3 \cdot 1 \cdot 1$

let

$$\left\langle \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}, \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} \right\rangle$$

$$= 4x_1x_2 + x_1y_2 + x_2y_1 + 3y_1y_2$$

1) Compute the length of $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

2) Compute the angle between $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$

3) Find a vector with length 1

$$\left\| \frac{1}{3} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\| = 1$$

$$\|kv\| = k\|v\|$$

$$\theta \approx 73.2^\circ$$

$$= 4 + 1 + 1 + 3 = 9$$

$$\left\| \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\| = \sqrt{9} = 3$$

$$2) \cos \theta = \frac{\left\langle \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\rangle}{\left\| \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\| \left\| \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\|}$$

$$\left\langle \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\rangle = 4 \cdot 1 \cdot 0 + 1 \cdot 1 + 0 \cdot 0 + 3 \cdot 0 \cdot 1 = 1$$

$$\left\| \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\| = \sqrt{4 \cdot 1 + 0 + 0 + 0} = 2$$

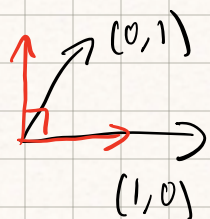
$$\left\| \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\| = \sqrt{0 + 0 + 0 + 3 \cdot 1} = \sqrt{3}$$

$$\cos \theta = \frac{1}{2\sqrt{3}}$$

$$\sqrt{\left\langle \begin{bmatrix} a \\ b \end{bmatrix}, \begin{bmatrix} a \\ b \end{bmatrix} \right\rangle} = 1$$

$$4a^2 + 2ab + 3b^2 = 1$$

formula for the unit circle relative to $\langle \cdot, \cdot \rangle$



in the red basis $\left\langle \begin{bmatrix} a_1 \\ b_1 \end{bmatrix}_R, \begin{bmatrix} a_2 \\ b_2 \end{bmatrix}_R \right\rangle$
 $= a_1a_2 + b_1b_2$

Ex fix a basis for $\mathbb{R}^3$ let $\left\langle \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix}, \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} \right\rangle$

$$= 2x_1x_2 + x_1y_2 + x_2y_1 + 3y_1y_2 + 2(y_1z_2 + y_2z_1) + 4z_1z_2$$

1) compute length of $\begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} = v$

$$\|v\| = \sqrt{2+2+2+3/4+2(2(-1)+2(-1))+4} = \sqrt{14}$$

2) compute the angle between $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$

$$\cos \theta = \frac{\left\langle \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\rangle}{\left\| \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\| \left\| \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\|} = \frac{(0+1+0+0+0+0)}{\left\| \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\| \left\| \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\|} = \frac{1}{\sqrt{2} \sqrt{3}} = \frac{\sqrt{6}}{6}$$

$$\theta = \cos^{-1}\left(\frac{\sqrt{6}}{6}\right) = \text{a bit more than } \frac{\pi}{3}$$

$$3) \frac{17}{\sqrt{14}} \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$$

Thm: given a finite ^{dim = n} vector space V and a fixed basis for V
1) in coordinates the formula for $\langle v, w \rangle$ is $w^T M v$

Where M is symmetric and has strictly positive eigenvalues

Ex $\langle \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}, \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} \rangle = 4x_1x_2 + x_1y_2 + x_2y_1 + 3y_1y_2$

$$M = \begin{bmatrix} 4 & 1 \\ 1 & 3 \end{bmatrix} \quad \begin{bmatrix} x_2 & y_2 \end{bmatrix} \begin{bmatrix} 4 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} x_2 & y_2 \end{bmatrix} \begin{bmatrix} 4x_1 + y_1 \\ x_1 + 3y_1 \end{bmatrix} = 4x_1x_2 + y_1x_2 + x_1y_2 + 3y_1y_2$$

2) given $n \times n$ M which is symmetric and has strictly positive eigenvalues $(v, w) \mapsto w^T M v$ defines an inner product